

A12 Trajectory Length and the Sangbun Integral

This document distinguishes two calculations: “A12 trajectory length” measures length; the “Sangbun integral” measures the area of the strip generated by the Sangbun vector field.

Part 1. A12 Trajectory Length | pp. 1-9

Find the distance traveled by the Sangbun-vector endpoint A12. The symbol $\mathbf{q}(s)$ denotes its position vector.

$$L = \int_4^{10} \|\mathbf{q}'(s)\| ds$$

$$L \approx 33.26167$$

Part 2. Sangbun Integral | pp. 10-13

Find the area of the strip swept out by the segments AA12 joining the base curve A to the trajectory of A12.

$$\mathcal{A} = \int_4^{10} \int_0^1 \|\mathbf{F}_s \times \mathbf{F}_u\| du ds$$

$$\mathcal{A} \approx 232.91171$$

Both values are numerical results for the same experimental 3D extension. Length and area have different units: units and square units. Overlapping strips are counted with multiplicity.

1. Dividing the Interval into n Parts

The interval is $s = 4$ to $s = 10$. The parameter s specifies position on the curve; it is not itself the actual arc length.

$$\Delta s = \frac{10 - 4}{n} = \frac{6}{n}$$

$$s_k = 4 + \frac{6k}{n}, \quad k = 0, 1, \dots, n$$

To make n segments, use $n + 1$ positions including both endpoints. The index k numbers these positions.

Segments n	Parameter step	Positions
2,000	0.003	2,001
4,000	0.0015	4,001
8,000	0.00075	8,001

For 2,000 segments

$$4, \quad 4.003, \quad 4.006, \quad \dots, \quad 10$$

Calculate A12 at each position. Equal increments of s do not necessarily give equal distances between adjacent A12 points.

2. A Three-Segment Example

Before using many segments, take $n = 3$ to see the procedure. The step is $6 \div 3 = 2$.

$$s_0 = 4, \quad s_1 = 6, \quad s_2 = 8, \quad s_3 = 10$$

Calculate $A12(4)$, $A12(6)$, $A12(8)$, and $A12(10)$, then add the three distances between consecutive points.

$$d_k = \|\mathbf{q}(s_k) - \mathbf{q}(s_{k-1})\|$$
$$L_3 = d_1 + d_2 + d_3$$

Connected positions	Straight-line distance
$s=4 \rightarrow s=6$	9.61247
$s=6 \rightarrow s=8$	11.05464
$s=8 \rightarrow s=10$	12.44168

$$L_3 \approx 33.10880$$

These values use the experimental 3D extension on page 7. The displayed distances are rounded; the total was calculated from unrounded coordinates.

3. The Full Coordinate Sum

$$\mathbf{q}(s) = (X(s), Y(s), Z(s))$$

Here X, Y, Z are the three coordinates of endpoint A12. Distinguish them from the normalized X, Y used earlier for the closed-form rotational components.

$$\Delta X_k = X(s_k) - X(s_{k-1})$$

$$\Delta Y_k = Y(s_k) - Y(s_{k-1})$$

$$\Delta Z_k = Z(s_k) - Z(s_{k-1})$$

$$L_n = \sum_{k=1}^n \sqrt{(\Delta X_k)^2 + (\Delta Y_k)^2 + (\Delta Z_k)^2}$$

Square the coordinate differences, add them, and take the square root to get the distance between two points in space. Sum over all segments.

For a planar curve, the Z -coordinate difference is zero; omit the Z -term.

Do not multiply by Δs again: this formula already sums distances. It differs from a Riemann sum of speeds.

4. From Partition Sums to an Integral

If $\mathbf{q}$ is continuously differentiable throughout the interval, the change in position over a small interval is approximated by:

$$\mathbf{q}(s_k) - \mathbf{q}(s_{k-1}) \approx \mathbf{q}'(s_{k-1})\Delta s$$

$$L_n \approx \sum_{k=1}^n \|\mathbf{q}'(s_{k-1})\| \Delta s$$

The right side is a Riemann sum: speed times a small parameter increment. As the increment tends to zero, it becomes the arc-length integral.

$$L = \lim_{n \rightarrow \infty} L_n = \int_4^{10} \|\mathbf{q}'(s)\| ds$$

$$L = \int_4^{10} \sqrt{X'(s)^2 + Y'(s)^2 + Z'(s)^2} ds$$

This is the standard arc-length theorem under the stated smoothness condition, not a conjecture inferred only from a numerical table.

If an antiderivative is known, evaluate it at the endpoints. Otherwise, approximate using partition sums or another numerical integration method.

5. Numerical Results and Their Limits

These results use finer partitions of the same 3D example. They use actual A12 coordinates, not arrows shortened for display.

Segments	Approximate A12 trajectory length
2,000	33.261665163
4,000	33.261665428
8,000	33.261665494

$$L \approx 33.26167$$

All three results round to the same five decimal places. This confirms numerical stability, but does not rigorously bound the error from the true value.

Why does the value increase slightly?

Adding a midpoint to a straight segment cannot decrease the sum of lengths, by the triangle inequality. Thus bisecting the same partition gives $L(2n) \geq L(n)$.

$$L_n \leq L_{2n} \leq L$$

This does not imply monotonic increase for every n . The relation holds for refinements that retain the existing points.

6. The 3D Rule Used in This Example

$$\mathbf{r}(s) = \left(s, \frac{s^2}{20} + 6, \frac{s^3}{200} \right)$$

$$4 \leq s \leq 10$$

This is not claimed as the unique definitive 3D definition. The calculation fixes the following experimental extension.

1) Take the plane containing O and the tangent at A. 2) Project the global X-axis orthogonally onto that plane to define its horizontal direction. 3) Apply the 2D A12 formula in that frame, then map back to space.

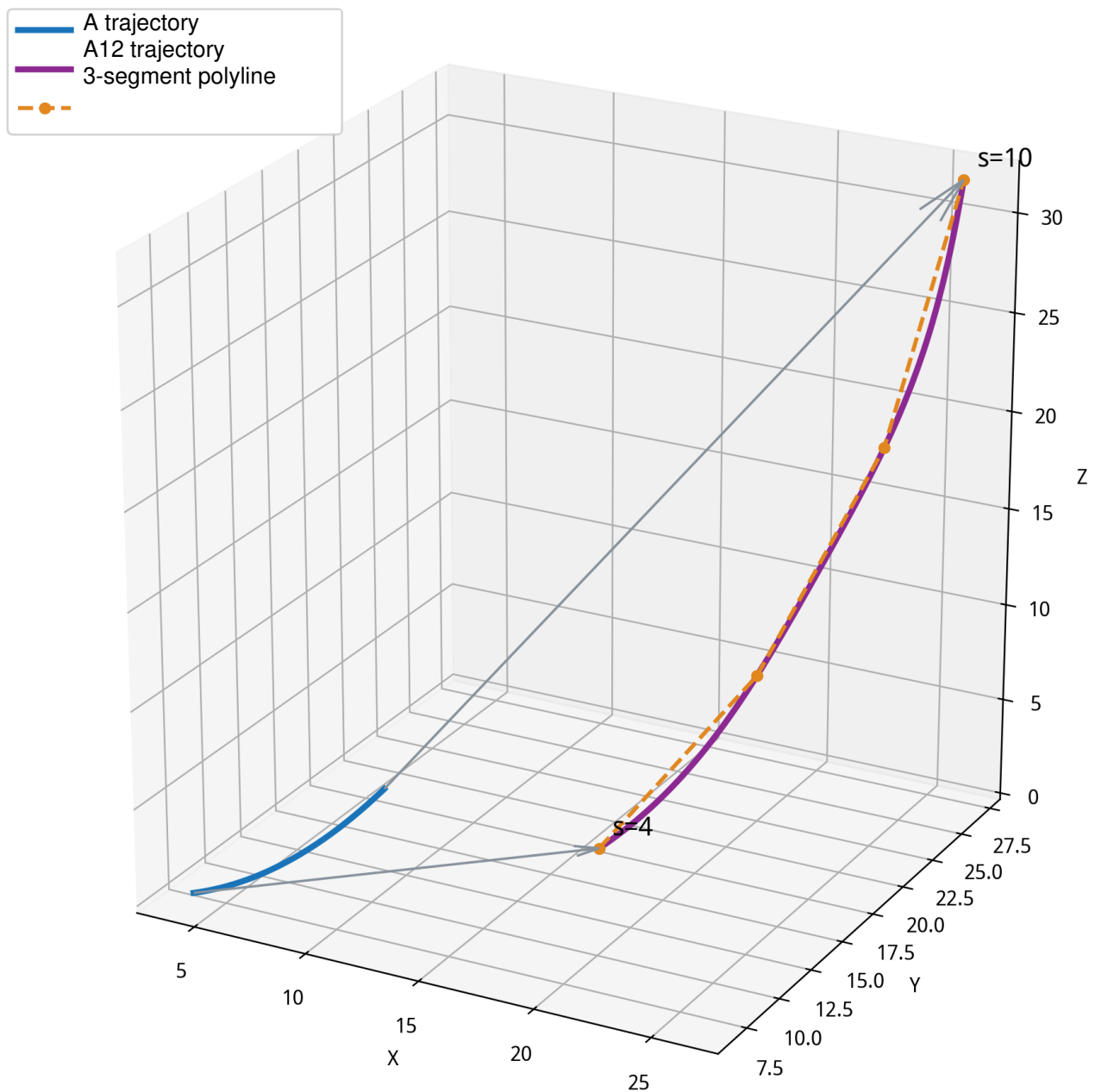
$$\mathbf{n} = \frac{\mathbf{r}' \times \mathbf{r}}{\|\mathbf{r}' \times \mathbf{r}\|}$$

$$\mathbf{e}_1 = \frac{\mathbf{e}_x - (\mathbf{e}_x \cdot \mathbf{n})\mathbf{n}}{\|\mathbf{e}_x - (\mathbf{e}_x \cdot \mathbf{n})\mathbf{n}\|}, \quad \mathbf{e}_2 = \mathbf{n} \times \mathbf{e}_1$$

Here $\mathbf{e}_x = (1, 0, 0)$. The planar coordinates of A are $(\mathbf{r} \cdot \mathbf{e}_1, \mathbf{r} \cdot \mathbf{e}_2)$, and the tangent slope is $(\mathbf{r}' \cdot \mathbf{e}_2)/(\mathbf{r}' \cdot \mathbf{e}_1)$. All numerical results here use this same coordinate choice.

Points where this frame is undefined, or a construction denominator is zero, need separate treatment. Changing the calculation rule can change the 3D results.

7. Visualizing the Endpoint Trajectory



Purple shows the trajectory of A12; the orange dashed line joins it with three straight segments. Finer partitions follow the purple curve more closely.

Split intervals at jumps. Integrating the derivative alone does not automatically include jump distances. At singularities, first check whether finite length exists.

8. The Essential Formula

For this interval, choose the desired number of segments n . In the formula below, q always means the actual position of A12.

$$L_n = \sum_{k=1}^n \left\| \mathbf{q}\left(4 + \frac{6k}{n}\right) - \mathbf{q}\left(4 + \frac{6(k-1)}{n}\right) \right\|$$

For a different interval

If the starting parameter is α and the ending parameter is β , with $\alpha < \beta$, use:

$$s_k = \alpha + \frac{(\beta - \alpha)k}{n}$$

$$L_n = \sum_{k=1}^n \left\| \mathbf{q}(s_k) - \mathbf{q}(s_{k-1}) \right\|$$

$$L = \lim_{n \rightarrow \infty} L_n$$

On an interval where the construction is defined and the trajectory is continuously differentiable, this limit gives its finite length.

Adding the lengths of the vectors AA12 is different from adding distances between consecutive A12 endpoints. Here L is the latter: endpoint trajectory length.

9. Definition of the Sangbun Integral

The Sangbun vector field consists of the vectors AA_{12} at points of the base curve. The Sangbun integral is defined here as the area of the strip filled by these vector segments.

$$\mathbf{S}(s) = \mathbf{q}(s) - \mathbf{r}(s)$$

$$\mathbf{F}(s, u) = \mathbf{r}(s) + u\mathbf{S}(s)$$

$$4 \leq s \leq 10, \quad 0 \leq u \leq 1$$

$\mathbf{r}(s)$ is the position of A on the base curve; $\mathbf{q}(s)$ is the position of A_{12} . $u = 0$ gives A , $u = 1$ gives A_{12} , and intermediate u fills the segment AA_{12} .

$$\mathcal{A} = \int_4^{10} \int_0^1 \|\mathbf{F}_s \times \mathbf{F}_u\| du ds$$

On smoothly defined intervals, apply the standard surface-area formula to sum the small areas of the strip. This area is called the Sangbun integral in this document.

Overlap is counted with multiplicity. The area of the union, counting overlapping regions only once, is a separate calculation.

10. Calculating the Sangbun Integral

$$\mathbf{F}_s = \mathbf{r}'(s) + u\mathbf{S}'(s)$$

$$\mathbf{F}_u = \mathbf{S}(s)$$

$$G(s, u) = \|(\mathbf{r}' + u\mathbf{S}') \times \mathbf{S}\|$$

$$\mathcal{A} = \int_4^{10} \int_0^1 G(s, u) du ds$$

Divide the s -interval into n parts and the u -interval into m parts to form small cells. Evaluating G at each cell center gives this approximation:

$$\mathcal{A}_{n,m}^{\text{mid}} = \frac{6}{nm} \sum_{i=1}^n \sum_{j=1}^m G(s_i^*, u_j^*)$$

$$s_i^* = 4 + \frac{6(i - 1/2)}{n}, \quad u_j^* = \frac{j - 1/2}{m}$$

The high-precision calculation used Gauss-Legendre quadrature with carefully chosen nodes and weights, instead of uniform midpoints. Results were compared with more sample points in each direction.

As an independent check, each cell was split into two triangles and their areas were summed.

11. Results and Cross-Checks

$$\mathcal{A} \approx 232.91171$$

Area is measured in the square of the input coordinate unit: square meters if coordinates are in meters, or simply square units if no unit is specified.

Method 1: Surface-area quadrature

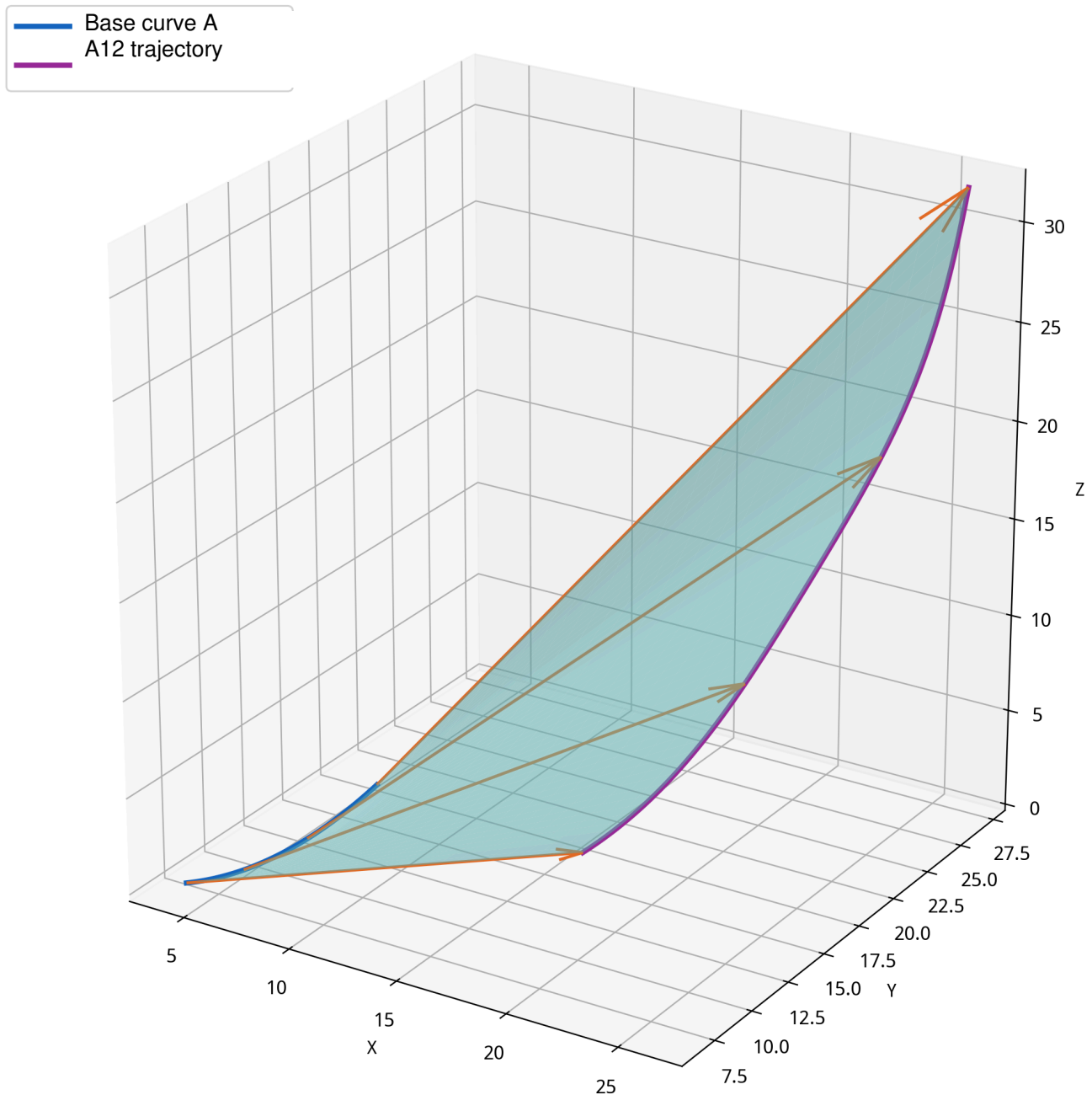
Samples in each direction	Approximate area
24 × 24	232.911700547
48 × 48	232.911707461
96 × 96	232.911707282
192 × 192	232.911707282

Method 2: Sum of Small Triangle Areas

Grid subdivisions	Approximate area
100 × 100	232.917260757
200 × 200	232.913095992
400 × 400	232.912054481
800 × 800	232.911794083

Both methods numerically approach the same value. This is not a certified error bound or a proof of an exact closed form. The extra digits are calculation outputs for comparison.

12. Sangbun Vector Field and Integral



Blue shows A, and purple shows the A12 trajectory. Orange arrows are the actual vectors $\vec{AA_{12}}$. The teal strip is the surface whose area is calculated. Vector lengths are not scaled down.

This is the experimental 3D extension on page 7 for $s = 4$ to 10. Changing the coordinate frame or the A12 rule can change the area.